S1023
GENERAL MATHEMATICS **III**
1 hour 45 minutes

Name:...

Examination Number:..

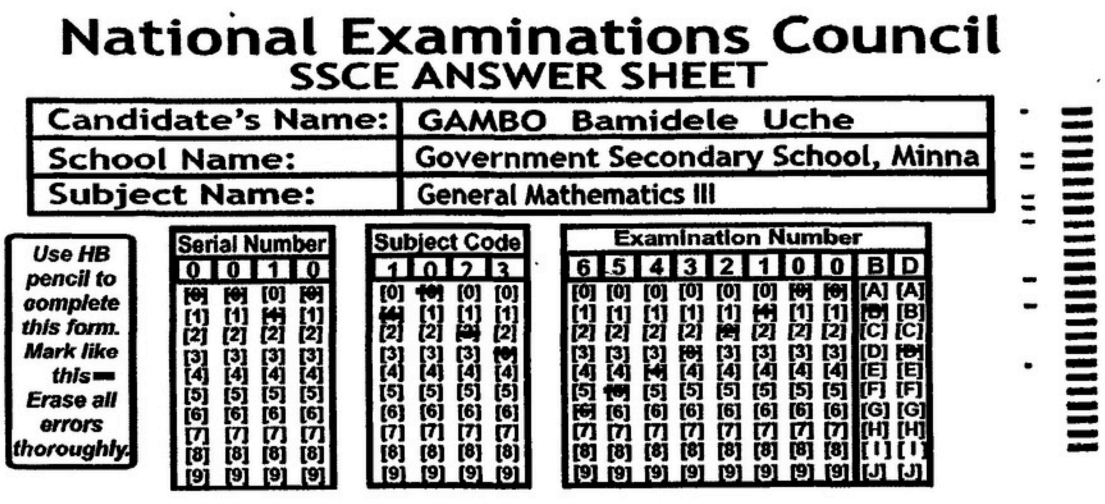
NATIONAL EXAMINATIONS COUNCIL
Senior School Certificate Examination

1 hour 45 minutes

GENERAL MATHEMATICS
PAPER III

Do **not** open this question booklet until you are told to do so. While waiting, read the following carefully.

1. Use **HB** pencil throughout.
2. **Use of mobile phone is not allowed.**
3. **Use of scientific calculator is allowed.**
4. **All diagrams are not drawn to scale.**
5. Take π to be $\frac{22}{7}$ **except** otherwise stated.
6. Where your answer sheet is **not** customised, provide the following information:
   (a) In the space marked *Candidate's Name,* write your **surname** in capital letters followed by your **other names.**
   (b) In the space marked *School Name,* write the name of your **school,** and in the space marked *Subject Name,* write **General Mathematics III.**
   (c) In the box marked *Subject Code*, write the digits **1023** in the spaces. There are numbered spaces in line with each digit. Shade carefully the space with the same number as each digit.
   (d) In the box marked *Examination Number*, write your **examination number** in the spaces at the top of the box. Shade the corresponding numbered spaces in the same way as for Subject Code.
7. An example is given below. This is for a candidate whose name is **GAMBO Bamidele Uche,** with **serial number 0010, examination number 65432100BD,** and who is taking **General Mathematics III (1023).**



Turn over

PAPER III

Answer **all** questions

Each question is followed by **five** options lettered **A – E.** Choose the correct option for each question and shade **in pencil** on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only **one** answer to each question and erase completely any answer you wish to change. Do all rough work on this question paper.

An example is given below.

The product of three numbers is 3876. Two of the numbers are 17 and 19. What is the third number?

A. 57
B. 12
C. 6
D. 3
E. 2

The correct option is '12' which is lettered **B.** Therefore answer space **B** would be shaded.

[A] [redacted] [C] [D] [E]

1. Express 0.001025 in standard form.

A. 1.025×10^{-5}
B. 1.025×10^{-4}
C. 1.025×10^{-3}
D. 1.025×10^{-2}
E. 1.025×10^{-1}

2. Find the positive difference between rounding off of 3,847,934,000 to the nearest billion and million.

A. 150,000,000
B. 152,000,000
C. 200,000,000
D. 250,000,000
E. 252,000,000

3. Evaluate $\dfrac{\left(\frac{1}{81}\right)^{-\frac{1}{2}}}{0.3} \times (0.09)^{\frac{1}{2}}$.

A. 81
B. 9
C. 3
D. $\frac{1}{3}$
E. $\frac{1}{9}$

4. Simplify $(0.027)^{\frac{1}{3}} \times (0.09)^{-\frac{3}{2}}$.

A. $\left(\frac{3}{10}\right)^{-2}$

B. $\left(\frac{9}{10}\right)^{-2}$

C. $\left(\frac{3}{10}\right)^{\frac{1}{2}}$

D. $\left(\frac{3}{10}\right)^{2}$

E. $\left(\frac{3}{10}\right)^{4}$

5. A particular bank paid 50k dividend per share. What would be the percentage yield for a customer who invested ₦5.00 per share?

A. 1
B. 2
C. 5
D. 10
E. 50

6. Simplify $\frac{1.6 \times 0.0084}{0.0048}$ and leave your answer in standard form.

A. 2.8×10^{-2}
B. 2.8×10^{-1}
C. 2.8×10^{0}
D. 2.8×10^{1}
E. 2.8×10^{2}

7. The cost of a motorbike is ₦110,000.00. If the value depreciates by 10% yearly, what will be its value at the beginning of the following year?

A. ₦11,000.00
B. ₦88,000.00
C. ₦99,000.00
D. ₦108,900.00
E. ₦121,000.00

8. A and B are sets such that $n(A) = 26$, $n(B) = 19$ and $n(A \cap B) = 11$. Find $n(A \cup B)$.

A. 30
B. 34
C. 37
D. 45
E. 56

9. If $2y^{\frac{1}{4}} - 5 = -1$, find the value of y.

A. $\frac{1}{16}$
B. $\frac{1}{8}$
C. $\frac{1}{2}$
D. 16
E. 81

10. Solve the equation $\left(2^{3x}\right)^{\frac{1}{2}} = \frac{1}{0.25}$.

A. $\frac{2}{3}$

B. $\frac{3}{4}$

C. $1\frac{1}{3}$

D. 2
E. 3

11. How many years will ₦100,000.00 amount to ₦180,000.00 at 5% simple interest per annum?

A. 4
B. 16
C. 20
D. 36
E. 50

12. If $B=\begin{pmatrix} 2 & 1 & 0 \\ 3 & 0 & -2 \\ 1 & -1 & 3 \end{pmatrix}$, determine the transpose of B.

A. $\begin{pmatrix} 2 & 3 & 1 \\ 1 & 0 & -1 \\ 0 & 2 & 3 \end{pmatrix}$

B. $\begin{pmatrix} -2 & 3 & 1 \\ 1 & 0 & -1 \\ 0 & -2 & 3 \end{pmatrix}$

C. $\begin{pmatrix} 2 & 3 & 1 \\ 1 & 0 & -1 \\ 0 & -2 & 3 \end{pmatrix}$

D. $\begin{pmatrix} 2 & -3 & 1 \\ 1 & 0 & -1 \\ 0 & -2 & 3 \end{pmatrix}$

E. $\begin{pmatrix} 1 & -1 & 3 \\ 3 & 0 & -2 \\ 2 & 0 & 1 \end{pmatrix}$

13. If y is a whole number such that $3y+4 \equiv 7 \bmod 8$, find the value of y.

A. 1
B. 2
C. 3
D. 4
E. 5

14. The 3rd and 7th terms of a Geometric Progression (G.P.) are 72 and $\frac{8}{9}$ respectively.
Find the 1st term.

A. 720
B. 648
C. 576
D. 288
E. 216

15. In a class of 30 students, every student has at least a ruler. If 25 students have white rulers and 20 students yellow rulers, how many students have rulers of both colours?

A. 55
B. 50
C. 45
D. 15
E. 10

16. Five numbers form a Geometric Progression (G.P.). If the 2nd and 5th numbers are 9 and 243 respectively, find the common ratio.

A. –9
B. –3
C. 2
D. 3
E. 5

17. The first and last terms of an Arithmetic Progression (A.P.) are $-2a$ and $38a$ respectively. Determine the number of terms if its common difference is $5a$.

A. 7
B. 8
C. 9
D. 8a
E. 9a

18. On Clara's tenth birthday, she began to save ₦2,000 .00 every year in a kidsave account with an annual interest rate of 20% compounded annually. What will be the worth of the account when she turns 18?

A. ₦8,599.63
B. ₦20,799.00
C. ₦32,998.17
D. ₦41,597.80
E. ₦46,000.00

19. The cost (C) of a car service is partly constant and partly varies with the time (T) it takes to do the work. If it costs ₦500.00 for a 2-hour service and ₦300.00 for an hour service, find the cost of a 3-hour service.

A. ₦600.00
B. ₦650.00
C. ₦680.00
D. ₦700.00
E. ₦800.00

20. Find the gradient of a line joining the points $(-5, -6)$ and $(5, -2)$.

A. $\frac{1}{4}$
B. $\frac{2}{5}$
C. $\frac{1}{2}$
D. $1\frac{1}{3}$
E. $1\frac{1}{2}$

21. Given that e: ABCD is a rhombus.
f: ABCD is a rectangle.
h: ABCD is a square.
Which of the following statements represents $e \wedge f \Leftrightarrow h$?

A. ABCD is a rectangle if it is a square and rhombus.
B. ABCD is a square if and only if it is a rhombus and rectangle.
C. ABCD is a square if and only if it is not a rhombus and rectangle.
D. ABCD is a rectangle if and only if it is not a square and rectangle.
E. ABCD is a rhombus if it is not a rectangle and square.

22. Solve the simultaneous equations
$2a + 3b = 5,\ 2a + b = 1.$

A. $a = \frac{-1}{2},\ b = -2$

B. $a = \frac{-1}{2},\ b = 2$

C. $a = \frac{1}{2}, b = -2$

D. $a = -1,\ b = 2$

E. $a = 2,\ b = \frac{1}{2}$

23. Solve the equation
$\frac{2x-1}{2} - \frac{3x-2}{3} = \frac{x}{2}.$

A. -3

B. $-\frac{7}{3}$

C. $-\frac{1}{3}$

D. $\frac{1}{3}$

E. 3

24. Determine the equation whose product and sum of the roots are –4.5 and –3.2 respectively.

A. $10x^2 - 32x - 45 = 0$

B. $10x^2 + 32x + 45 = 0$

C. $10x^2 + 32x - 45 = 0$

D. $10x^2 + 16x - 9 = 0$

E. $10x^2 - 16x - 9 = 0$

25. If p: x is an even number.
q: x is a multiple of 3.
What does $\sim(p \wedge q)$ stand for?

A. x is an even number and a multiple of 3

B. x is an even number or a multiple of 3

C. x is not a multiple of 3 and is even

D. It is false that x is even and a multiple of 3

E. x is not an even number and not a multiple of 3

26. Solve the inequality $x < \frac{x}{3} + 4$.

A. $x < 12$

B. $x < 6$

C. $x < 4$

D. $x < 3$

E. $x < -12$

27. Simplify $\frac{a^2 - 4a + 3}{a^2 - a - 6}$.

A. $\frac{a-1}{a+2}$

B. $\frac{a+1}{a-2}$

C. $\frac{1-a}{a+2}$

D. $\frac{a-1}{a-2}$

E. $\frac{a+1}{a+2}$

28. The roots of a quadratic equation are -2 and $\frac{1}{3}$. Find the equation.

A. $3x^2+5x+2=0$
B. $3x^2-5x+3=0$
C. $3x^2-5x-2=0$
D. $3x^2+5x-2=0$
E. $3x^2-5x-3=0$

29. If $x \propto \frac{1}{y^2}$ and $x=4$ when $y=3$, find the value of x when $y=2$.

A. 18
B. 9
C. 4
D. 3
E. 1

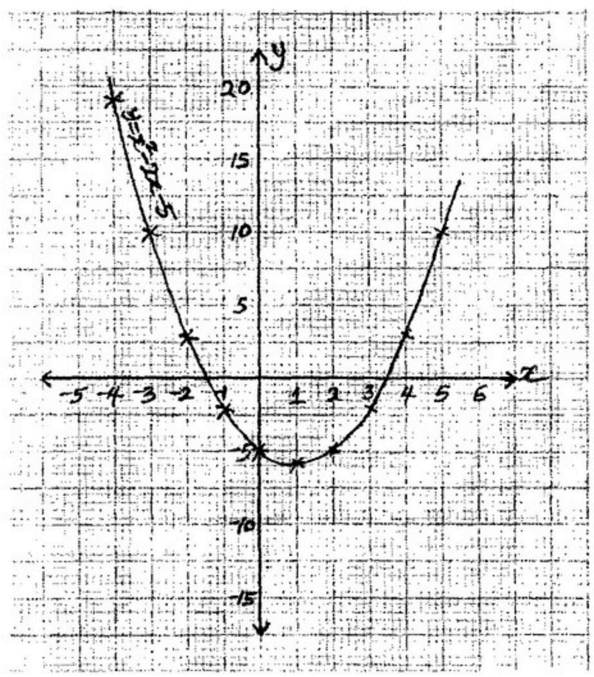
30. Make b the subject of the formula $d=\sqrt{\frac{ab}{c}-c^2b}$.

A. $b=\frac{cd^2}{a-c^3}$
B. $b=\frac{d^2}{a-c^2}$
C. $b=\frac{d^2}{c^2-a}$
D. $b=\frac{cd}{a-c^3}$
E. $b=\frac{a-c^3}{cd^2}$

31. If u: I like bread with butter.
v: I like lemon for tea.
Translate "I like lemon for tea, but not bread with butter" into symbol.

A. $u \wedge v$
B. $v \vee u$
C. $v \wedge \sim u$
D. $\sim u \vee v$
E. $u \vee \sim v$

Use the graph below to answer questions **32** and **33.**



32. Determine the solutions of the curve.

A. $x=-1.2$ *or* 3.2
B. $x=-1.3$ *or* 3.4
C. $x=-1.4$ *or* 3.5
D. $x=-1.5$ *or* 4.0
E. $x=-1.5$ *or* 5.0

33. Find the least value of y.

A. -5.2
B. -5.1
C. -4.5
D. -1.4
E. 3.5

34. If $\cos\alpha = \frac{3}{5}$ and $0^o < \alpha < 90^o$, evaluate $3\cos\alpha - 2\sin\alpha$.

A. $\frac{1}{5}$
B. $\frac{2}{5}$
C. $\frac{1}{2}$
D. $\frac{3}{5}$
E. $\frac{4}{5}$

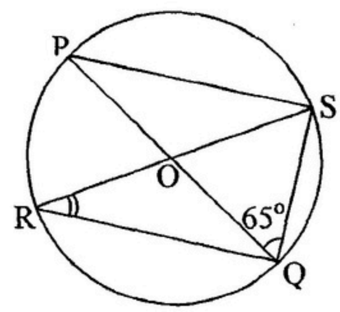
35. The bearing of Y and Z from X are 061^o and 110^o respectively. Calculate |YZ| if |YX| = 3 cm and |XZ| = 5 cm, correct your answer to 2 significant figures.

A. 3.7 cm
B. 3.8 cm
C. 4.0 cm
D. 4.1 cm
E. 5.0 cm

36. The curved surface area of a hemisphere is 7,700 cm^2. Find its diameter.
(Take $\pi = \frac{22}{7}$)

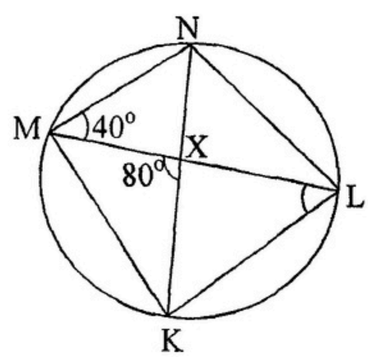
A. 17.5 cm
B. 35.0 cm
C. 70.0 cm
D. 70.5 cm
E. 87.7 cm

37. In the figure below, $\overline{PQ}$ is a diameter of the circle PSQR with centre O. If $P\hat{Q}S = 65^o$, find $Q\hat{R}S$.



A. 115^o
B. 90^o
C. 65^o
D. 40^o
E. 25^o

38. In the diagram below, MKLN is a cyclic quadrilateral and the diagonals ML and KN intersect at X.
If $N\hat{M}L = 40^o$ and $M\hat{X}K = 80^o$, calculate $M\hat{L}K$.



A. 100^o
B. 80^o
C. 60^o
D. 40^o
E. 20^o

39. Find the y-intercept of the line $4y + 3x - 2 = 0$.

A. -2
B. $-\frac{3}{4}$
C. $-\frac{1}{2}$
D. $\frac{1}{2}$
E. 2

40. The angle of a sector of a circle is 295°. If the area of the sector is 35.2 cm^2, calculate its radius to 2 decimal places.
(Take $\pi = \frac{22}{7}$)

A. 2.10 cm
B. 3.45 cm
C. 3.70 cm
D. 4.31 cm
E. 5.71 cm

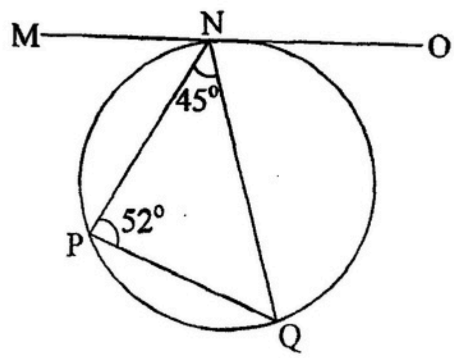
41. Find the angle between two straight lines whose slopes are 4 and 5, correct to 1 d.p.

A. 2.5°
B. 2.6°
C. 2.7°
D. 2.8°
E. 2.9°

42. If the radius of a parallel of latitude is 4,500 km, find the latitude correct to the nearest degree.
(Take R = 6,400 km)

A. 45° East or West
B. 45° North or East
C. 45° North or South
D. 50° East or West
E. 50° North or South

43. In the diagram below $\overline{MNO}$ is a tangent to the circle. Calculate $P\hat{N}O$.



A. 135°
B. 128°
C. 97°
D. 83°
E. 77°

44. A sphere has a radius of 7 cm. Find its surface area.

A. 616 cm^2
B. 206 cm^2
C. 205 cm^2
D. 154 cm^2
E. 88 cm^2

45. Calculate the perimeter of a sector whose radius is 21 cm and its angle at the centre is 150°.

A. 2310 cm
B. 1310 cm
C. 210 cm
D. 105 cm
E. 97 cm

46. Find the equation of a line that passes through the point (1, 2) whose gradient is $\frac{1}{3}$.

A. $3y + x = 7$
B. $3y - x = -5$
C. $3y - x = 7$
D. $3y - x = 5$
E. $y - 3x = 7$

47. Calculate the area of a circle of radius 3.69 cm, leaving your answer in 2 significant figures.

(Take $\pi = \frac{22}{7}$)

A. 11 cm^2
B. 12 cm^2
C. 23 cm^2
D. 42 cm^2
E. 43 cm^2

48. Calculate the circumference of the parallel of latitude 60°S correct to 3 significant figures.

(Take R = 6,400 km and $\pi = \frac{22}{7}$)

A. 20,000 km
B. 20,100 km
C. 20,110 km
D. 20,114 km
E. 40,200 km

49. Find the mean score of ten students if the variance of their scores is 8 and $\sum x^2 = 2040$.

A. 2.82
B. 11.83
C. 14.00
D. 45.14
E. 64.00

50. The mean of five numbers is 6. If the numbers are 1, 6, x, 8, 7, what is the mode?

A. 1
B. 5
C. 6
D. 7
E. 8

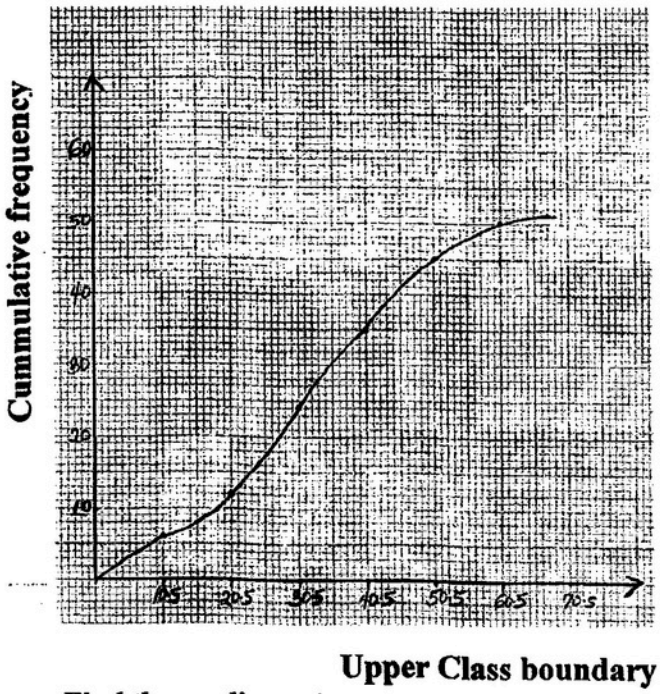
51. Which of the following is **not** a measure of dispersion?

A. mean deviation
B. median
C. percentile
D. standard deviation
E. variance

52. The standard deviation of the ages of 10 students is 1.2. What is the variance?

A. 3.46
B. 3.16
C. 2.80
D. 1.86
E. 1.44

Use the cumulative frequency curve below to answer questions **53 to 55.**



53. Find the median value.

A. 50.0
B. 34.5
C. 30.5
D. 25.0
E. 10.0

54. Calculate the 6th decile.

A. 50.0
B. 40.0
C. 45.5
D. 36.0
E. 20.0

55. Determine the 80th percentile.

A. 60.5
B. 55.5
C. 45.5
D. 40.5
E. 35.5

Use the information below to answer questions **56** and **57.**

The table below shows the age distribution of students in a school choir.

Age	12	13	14	15	16	17
Frequency	2	1	3	6	5	3

56. How many students are in the school choir?

A. 10
B. 15
C. 20
D. 25
E. 30

57. What is the modal age?

A. 6
B. 8
C. 10
D. 15
E. 17

58. Differentiate $3x^2 - 7x - 4$ with respect to x.

A. $3x^3 - 7x^2 - 4$
B. $x^3 - \frac{7}{2}x^2 - 4x$
C. $6x^2 - 7$
D. $6x - 7$
E. $7x - 4$

59. If $y = x(x^3 - x + 1)$, evaluate $\int_{-1}^{1} y\,dx$.

A. $-\frac{4}{15}$
B. 0
C. $\frac{11}{15}$
D. 1
E. 3

60. What is the value of $\int_0^1 (1-2x)^2\,dx$?

A. $-\frac{1}{8}$
B. $-\frac{1}{4}$
C. 0
D. $\frac{1}{3}$
E. 4

S1022 **GENERAL** **MATHEMATICS** **II** **2 hours 30 minutes**

Name:...

Examination Number:...

NATIONAL EXAMINATIONS COUNCIL
Senior School Certificate Examination

2 hours 30 minutes

GENERAL MATHEMATICS
PAPER II

Do **not** open this question booklet until you are told to do so. While waiting, read the following carefully.

Write your **Name** and **Examination Number** in the spaces provided at the top right-hand corner of this question booklet.

The paper is in two parts: **I** and **II**, and will last for **2 hours 30 minutes.**

Answer **all** the questions in Part **I** and **five** questions in Part **II**.

Write your answer in blue or black ink in your answer booklet.

You are to give your answers as correctly as the data and tables allow.

Answers are to be written in your answer booklet.

Use of mobile phone is not allowed.

Use of scientific calculator is allowed.

PAPER II

PART I

Attempt **all** questions in this part.

1(a) Without using mathematical tables, find the value of $\log_{10} 6 + \log_{10} 45 - \log_{10} 27$.

(b) Solve the equation $8^x = 32$.

(c) Simplify $\frac{81^{\frac{1}{6}}}{27^{-\frac{1}{9}}}$. **(8marks)**

2(a) A straight line passes through the points A(3, 0) and B(2, –1). Find the:
(i) Gradient of the line
(ii) Equation of the line

(b) ABCD is a parallelogram with /AD/=8 cm, /AB/ =6 cm and $A\hat{B}C = 60°$.
Calculate, correct to 3 significant figures, the area of:
(i) Triangle ABC
(ii) Parallelogram ABCD **(8marks)**

3(a) A village P is 12 km from village Q. It takes 3 hours 20 minutes to travel from Q to P and back to Q by a boat. If the boat travels at a speed of 6 km/h from P to Q and $(6 + x)$ km/h back to P, find the value of x.

(b) Find the quadratic equation whose roots are $\frac{2}{3}$ and $\frac{3}{4}$. **(8marks)**

4(a) Differentiate $y = \frac{x^2}{1+x^2}$ with respect to x.

(b) Solve the equation $\frac{2}{3}(3x+2) = \frac{3}{4}(2x-3)$. **(8marks)**

5(a) The table below shows the mass of logs of wood to the nearest kilogramme.

Mass (kg)	31 – 40	41 – 50	51 – 60	61 – 70	71 – 80	81 –90
Frequency	3	10	15	12	6	4

Calculate the:
(i) Mean
(ii) Mode **(8marks)**

PART II

Attempt **five** questions only in this part.

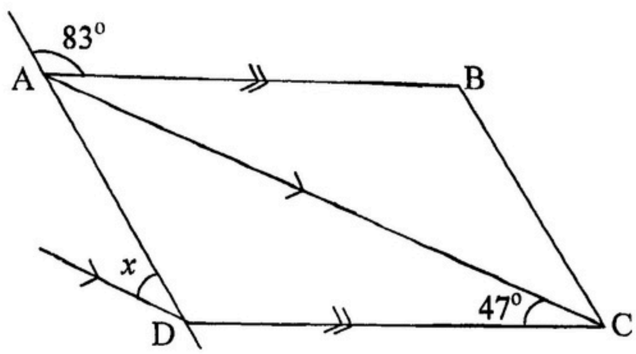
6(a) If the 2nd term of a Geometric Progression (G.P.) is 6 and the 4th term is 54, find the:

(i) 1st term
(ii) 5th term

(b) In a school of 160 pupils, 75 have pencils, 87 have pens and 93 have rulers. 25 have both pencils and pens, 30 have both pencils and rulers while 47 have both pens and rulers. Each pupil has at least one of the three items.

(i) Draw a venn diagram to illustrate this information.
(ii) How many pupils have pencils only? **(12marks)**

7(a) Use the information in the figure below to find the value of x.



(b) A boat sails 6 km on a bearing of 040° and then 8 km on a bearing of 100°.

(i) How far is the boat from its starting point?
(ii) Calculate the bearing of the boat from its starting point.
(Leave your answers to the nearest km and degree) **(12marks)**

8(a) Find the area enclosed by the curve $y = 2x^2 + 3$, the x – axis and the ordinates $x = 2$, $x = 1$.

(b) In a community, men contribute ₦p each and women ₦q each towards the Community Development Fund. In a week, 3 men and 5 women contributed a total of ₦9,500.00. In another week, 5 men and 10 women contributed a total of ₦17,500.00. Find the:

(i) Value of p and q
(ii) Total amount that will be contributed by 8 men and 12 women

(12marks)

9. An aircraft flies from P(12°N, 11°W) to Q(12°N, 11°E) in 2 hours and then changes its course and arrives at S(44°N, 11°E) in 6 hours after leaving Q. Calculate, correct to 3 significant figures, the:

(a) Total distance covered
(b) Average speed of the aircraft
(c) Time difference between Q and S
(Use $\pi = 3.142$ and R = 6,400 km) **(12marks)**

10(a) Copy and complete the table below for the relation $y = x^2 - 2x - 3$.

x	-3	-2	-1	0	1	2	3	4	5
y	12			-3				5	

(b) Draw the graph of the relation in (a) above using a scale of 2 cm to 1 unit on x –axis and 1cm to 1unit on y–axis. Hence, determine the roots of $x^2 - 2x - 3 = 0$.

(c) Using the same axis, draw the graph of $y = 2 - 2x$.

(d) Deduce the roots of the equation $x^2 - 2x - 3 = 2 - 2x$.

(e) Find the minimum value of $y = x^2 - 2x - 3$. **(12marks)**

11(a) Find the inverse of the matrix A where A $= \begin{pmatrix} 6 & 11 \\ 1 & 2 \end{pmatrix}$.

Hence, solve the matrix equation AX = B where X $= \begin{pmatrix} x \\ y \end{pmatrix}$ and B $= \begin{pmatrix} 29 \\ 5 \end{pmatrix}$.

(b) The first and last terms of an Arithmetic Progression (A. P.) are 5 and 89 respectively. If the sum of the A. P. is 611, find the:
(i) Number of terms in the A. P.
(ii) Common difference **(12marks)**

.12. The scores of 30 students in a Biology test are given below:

40	74	12	27	34	26
32	77	14	20	11	31
51	52	19	65	35	30
19	50	21	38	43	24
52	64	10	27	90	32

(a) (i) Construct a frequency table with class interval $1-20,\ 21-40\ldots$

(ii) Using an assumed mean of 30.5, calculate the standard deviation of the distribution.

(b) Find the probability that a student selected at random scored above 60 marks.

(12marks)